Yixiu Xiao

PhD Candidate in Mathematics, SJTU
Visiting Research Student, UNSW Sydney

Personal details

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Education

Apr 2025–present	**Visiting Research Student (Joint Training Program)**, *The University of New South Wales (UNSW Sydney)*, Sydney, Australia Supervisor: Prof. Igor E. Shparlinski.
Sep 2021–expected Mar 2027	**Ph.D. Candidate, Mathematics**, *Shanghai Jiao Tong University*, Shanghai, China Supervisor: Prof. Hongze Li.
Sep 2017–Jun 2021	**B.Sc. in Mathematics and Applied Mathematics**, *Northwest University*, Xi'an, China

Research Interests

Analytic Number Theory

- Exponential sums, character sums, and bilinear forms
- Distribution problems related to primes and modular arithmetic
- Kloosterman sums, Gauss sums, and trace functions
- Additive combinatorics and harmonic analysis in number theory
- Probabilistic aspects of analytic number theory

Current Research Directions

- Fractional parts of sums of square roots
- Bilinear sums involving Kloosterman and Gauss sums
- Shifted bilinear sums of Salié sums and the distribution of modular square roots of shifted primes
- Large sieve inequalities over short intervals
- Möbius-related exponential sums via the delta circle method

- Square-free parts of palindromes

Publications

Published

Least Zero of a Cubic Form, with Hongze Li. *Frontiers of Mathematics*, 2025. DOI: 10.1007/s11464-024-0113-6.

Least Zero of Pairs of Additive Cubic Equations, with Hongze Li. *Forum Mathematicum*, 2026. DOI: 10.1515/forum-2025-0367.

Preprints (arXiv)

Large sieve inequality for sums of Legendre symbols over short intervals, with Igor E. Shparlinski. *arXiv*: 2604.23661.

Shifted bilinear sums of Salié sums and the distribution of modular square roots of shifted primes, with Igor E. Shparlinski. *arXiv*: 2601.10113.

Scholarships

2024 China Scholarship Council (CSC), State Scholarship Fund (Visiting PhD Student to Australia), 24 months. Covers international airfare, living stipend, and health insurance.

Talks

Diophantine equations and cubic forms, Prime Pals / Number Theory Group Seminar (internal), UNSW Canberra, Canberra, 9 Sep 2025.

Least zero of pairs of additive cubic equations, AustMS Annual Meeting, La Trobe University, Melbourne, 10 Dec 2025.

Research Experience

Research Assistant, Prof. Hongze Li, Shanghai Jiao Tong University, 2021–present.

Conducted research in analytic number theory, including exponential sums, character sums, Diophantine equations, and related problems in number theory. Participated in literature study, mathematical discussions, proof development, and manuscript preparation.

Research Assistant, School of Mathematics and Statistics, UNSW Sydney, 7 April 2026–present. Supervisor: Prof. Igor E. Shparlinski.

Teaching

Mathematical Analysis, Teaching Assistant, Shanghai Jiao Tong University, Fall terms (2021–2023).

Elementary Number Theory (Summer Term), Teaching Assistant, Shanghai Jiao Tong University, 2022–2024.

MATH8000 Student Support (Student Supervision), UNSW Sydney, Term 3 2025.

MATH8000-Student Support (CTP), UNSW Sydney, Term 1 2026–present.

MATH1141 Higher Mathematics 1A, Routine Marker, UNSW Sydney, Term 1 2026–present.